FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. AC

Sol. For first refraction

$$\frac{\mu_2}{V}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$$

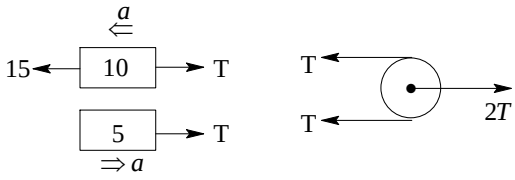
$$\frac{2}{v}-\frac{1}{\alpha}=\frac{2-1}{R}\Rightarrow v=2R$$

As 45° is greater than critical angle of 30° so the rays gets totally reflected at the second face. So the plane surface will behave like plane mirror.

2. ACD

Sol. $15-T=10a$

$$T=5a$$



So, $a=1m/s^2$

and $T=5N$

So, $2T=10N$

3. BC

Sol. As the plank retards 1kg block will move relative to the plank but 2kg block will remain at rest distance travelled by 1kg block in 1 sec.

$S_r = \frac{1}{2} \times a_r \times t^2 = \frac{1}{2} m$

Relative velocity of 1kg after 1 sec $V_r = 1 m/s$

Then using conservation of linear momentum and Newtons experimental Law

$V_1 + 2V_2 = 1 ----(1)$

$V_1 - V_2 = -\frac{1}{2} ----(2)$

$\Rightarrow V_1 = 0 \ \& \ V_2 = \frac{1}{2} \text{m/s}$

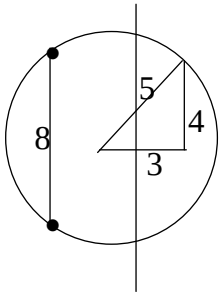
Final separation between blocks $= \frac{(1/2)^2}{2 \times 2} = \frac{1}{16} m$

4. AD

5. AC

6. AB

Sol. $16 \times 10^{-3} = 2T \times l$



$l = \frac{16 \times 10^{-3}}{2 \times 0.1}$

$= 8cm$

$\Rightarrow$ distance from centre = 3cm

$\Rightarrow 5 + 3 \ or \ 5 - 3cm$

7. ACD

8. ABD

Sol. Degree of freedom of He = 3

Degree of freedom of H_2 = 5

Average degree of freedom $= \frac{3 \times 2 + 5 \times 2}{2 + 2} = 4$

Now $\gamma = 1 + \frac{2}{f} = \frac{3}{2}$

For adiabatic process $\Delta Q = 0$ and $PV^{\gamma} = \text{constant}$ or $TV^{\gamma - 1} = \text{constant}$

From first law $\Delta U = \Delta W \qquad [\because \Delta Q = 0]$

9. A

10. C

11. A

12. C

SECTION – C

13. 00022.00

Sol. Motion will start when $F \geq \mu_s mg$ at $t = 1s$

After that $F - \mu_k mg = ma$

$3t^2 - 2 = a$ *or* $3t^2 - 2 = \dfrac{dv}{dt}$

$\Rightarrow v = \int_1^3 \left(3t^2 - 2\right)dt = 22\, m/s$

14. 00128.00

Sol. $\theta = \dfrac{14 \times 10^8}{1.5 \times 10^{11}} = \dfrac{14}{1.5} \times 10^{-3}$

$\omega = \dfrac{2\pi}{24 \times 3600}$

$t = \dfrac{\theta}{\omega}$

$t = \dfrac{14}{1.5} \times 10^{-3} \times \dfrac{24 \times 3600}{2\pi}$

$t = \dfrac{14 \times 8 \times 3.6}{\pi} \simeq 128\,\text{sec}$

15. 00017.00

Sol. $0 = 20 - 10t' \Rightarrow t' = 2\,\text{sec}$

$45 = 5t''^2 \Rightarrow t'' = 3$

So, $t_1 = 2 + 5 \times 3 = 17\,\text{sec}$

16. 00002.00

17. 00040.00

Sol. Maximum acceleration $= \omega^2 A = g$

$$A = \frac{g}{\omega^2} = 40\, cm$$

18. 00464.00

Sol. $75000 \times 10^{-10} \times T_{\max} = 3 \times 10^{-3}$

$\Rightarrow T_{\max} = 400\, K = T_B$

$\dfrac{T_A - 400}{\dfrac{1}{K} \times \dfrac{l}{A}} = \sigma A T_B^4 \quad \Rightarrow T_A = 464K$

Chemistry

PART – II

SECTION – A

19. B

20. CD

Sol. Electrolysis of molten LiH only gives H_2 at anode and the other hydrides tend to decompose before they melt. NaH has crystal structure similar to $NaCl$

21. BCD

Sol. (B) PV = nRT

$$P = \frac{nRT}{V} \therefore \ln p = \ln nRT - \ln V$$

$$\Rightarrow \log_{10} P = \log_{10}(nRT) - \log_{10}(V)$$

$\therefore$ slope = – 1

Intercept value = ln(nRT)

$$= \ln\left(1 \times 0.0821 \times \frac{1}{0.0821}\right) = 0$$

(C) PM = dRT

$$\frac{p}{d} = \frac{RT}{m} = \frac{1}{m} = \frac{1}{100} = 0.01$$

(D) PV^2 = nRTV

Slope = nRT $\Rightarrow$ 1

22. ABCD

Sol. Calculation of mol. wt. of BOH:

eq. of the base = eq. of H_2SO_4 used

$$\frac{0.496}{m_B} \times 1 = 40 \times 10^{-3} \times \frac{1}{2} \times 2 \Rightarrow m_B = 12.4 \text{ gmol}^{-1}$$

Now:

$$\Delta T_f = i \times K_f \times m$$

$$0.165 = i \times 1.86 \times \frac{1.5}{12.4} \times \frac{1000}{150}$$

$$i = \frac{0.165 \times 12.4}{1.86} = 1 - 1 = 1 + \alpha \therefore \alpha = 0.1$$

So, $(OH^-) = C\alpha = 0.08 \times 0.1 = 8 \times 10^{-3}$ $\therefore$ pH = 11.903

So, $K_b = C\alpha^2 = 0.8 \times (0.1)^2 = 8 \times 10^{-4}$

Also, $\pi = iCRT$

$= 1.1 \times 0.08 \times 0.0821 \times 300$

= 2.167 atm

23. BD

Sol. $3CS_2 + 6\ NaOH \longrightarrow Na_2CO_3 + 2Na_2CS_3 + 3H_2O$

24. ACD

Sol. From the given expression, $\Delta G° = +4.73 \text{ KJ/mol}$ & RT = + 2.5 KJ/mol.

This gives T = 300 K

Initially, $P_{NO_2} \to 0$ hence $\frac{P_{NO_2}^2}{P_{N_2O_4}} \to 0$ & $\ln\left[\frac{P_{NO_2}^2}{P_{N_2O_4}}\right] \to -\infty$

Hence ΔG expression is (–)ve & reaction moves in forward direction spontaneously.

Mole fraction of N_2O_4 is much larger than that of NO_2. Thus $P_{N_2O_4} > P_{NO_2}$

But for $\frac{P_{NO_2}^2}{P_{N_2O_4}}$ we can't say, hence comparison of K_p value = + antilog $\left[\frac{-\Delta G^0}{RT}\right] = 1.909$ with $Q_P = \frac{P_{NO_2}^2}{P_{N_2O_4}}$ is inclusive. Hence $Q_P > K_P$ or $Q_P < K_P$ or even $Q_P = K_P$ can be possible

25. ABCD

Sol. $X \to Sn(OH)Cl$

$Y \to Sn(OH)_2$

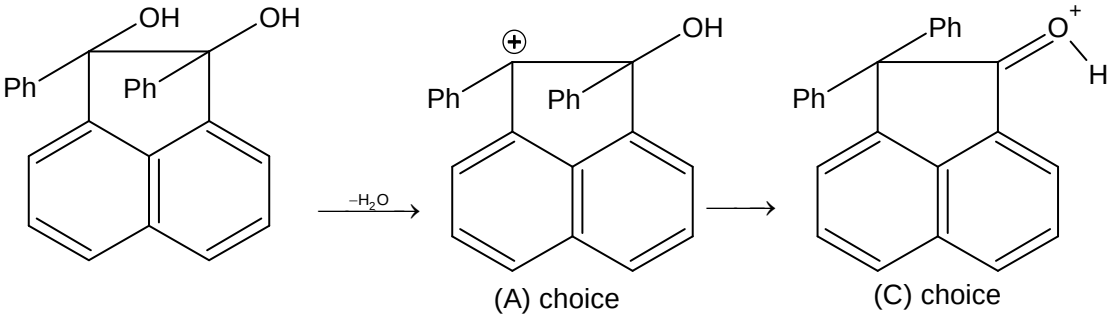
$Z \to Na_2SnO_2$

$P \to Au$

$Q \to HI$

26. AC

Sol.



(A) choice (C) choice

Hence (A) and (C) are correct while (B) and (D) are incorrect.

27. D

28. C

Sol. (for Q. 27-28)
A – (i), (ii), (iii)
B – (iv), (vi)
C – (i), (iii), (ix)
D – (iii), (vii)

29. C

30. C

Sol. (for Q. 29-30)
A – (ii, iii, v)
B – (I, iv)
C – (vi, vii)
D – (ii, iii)

SECTION – C

31. 00000.09

Sol. $\underset{1}{PCl_5} \rightleftharpoons \underset{0}{PCl_3} + \underset{0}{Cl_2}$

$\underset{1-\alpha}{\quad} \quad \underset{\alpha}{\quad} \quad \underset{\alpha}{\quad}$

$1+\alpha = \frac{D}{d}$

$1+\alpha = \frac{250.2}{208.5}$

$\alpha = 0.2$

Mole of $PCl_5 = 0.8$, PCl_3 $Cl_2 = 0.2$

$\therefore K = \frac{0.2 \times 0.2}{0.8} = \frac{1}{20}$

Now, when it transferred to 2L vessel reaction proceeds in forward direction.

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

$0.8-x \quad 0.2+x \quad 0.2+x$

$\frac{(0.2+x)^2}{(0.8-x)\times 2} = \frac{1}{20}$

$\therefore x = 0.07$

Degree of dissociation $= \frac{0.07}{0.08} = 0.0875(\text{approx}) = 0.09$

32. 00021.00

Sol. According to the law of conservation of momentum, momentum lost by photons = momentum gained by glow-worm

Momentum lost by photons = no of photons x momentum lost by one photon

$= \frac{\text{Total energy}}{\text{energy of 1 photon}} \times \frac{h}{\lambda} \left[\text{Debroglie' equation } \lambda = \frac{h}{p}.p = \frac{h}{\lambda}\right]$

$= \frac{\text{power} \times \text{time}}{\frac{hc}{\lambda}} \times \frac{h}{\lambda}$

$= \frac{\lambda}{hc} \times \text{power} \times \text{time} \times \frac{h}{\lambda}$

$= \frac{0.1 \times 365 \times 24 \times 60 \times 60 \times 10}{3 \times 10^8} \Rightarrow (1)$

Momentum gained by glow-worm $= \frac{5}{1000} \times (v) \text{ in m/sec} \longrightarrow (2)$

(1) = (2)

Hence v = 21 m/sec

Ans. $\frac{v}{7} = 3$.

33. 00006.90

Sol. $10Al + 6KMnO_4 + 27H_2SO_4 \longrightarrow 6KHSO_4 + 5Al_2(SO_4)_3 + 6MnSO_4 + 24H_2O$

34. 00001.75

Sol: H_2SO_4 produced = $\frac{2.463\times 0.01}{2\times 0.0821\times 300} = \frac{10^{-2}}{2}M$

$2NO_2 + H_2O \rightleftharpoons HNO_2 + HNO_3$

Conc. of $NO_2 = \frac{10^{-2}}{2}M$

So $HNO_2 = \frac{10^{-2}}{2}M$

$HNO_3 = \frac{10^{-2}}{2}M$

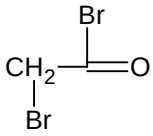
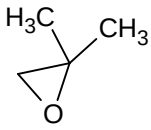
Net $\left[H^+\right] = \frac{10^{-2}}{2}\times 2 + \frac{10^{-2}}{2} + \frac{10^{-2}}{2}\times\frac{60}{100} = 1.8\times 10^{-2}$

pH = 1.75

35. 00357.90

Sol. One mole of compound consume 6 moles of CH_3MgBr so half mole will consume 3 moles of CH_3MgBr

36. 00072.00

Sol. A is  and B is 

Mathematics

PART – III

SECTION – A

37. AB

Sol. $S_m = \sum_{r=0}^{n} {}^nC_r \left[-\left(\frac{2^m - 1}{2^m} \right) \right]^r$

$S_m = \left[1 - \left(\frac{2^m - 1}{2^m} \right) \right]^n$

$S_m = \left(\frac{1}{2^m} \right)^n$

$\therefore \sum_{k=1}^{4} S_k = \left(\frac{1}{2} \right)^n + \left(\frac{1}{4} \right)^n + \left(\frac{1}{8} \right)^n + \left(\frac{1}{16} \right)^n$

$= \frac{\left(\frac{1}{2} \right)^n \left[1 - \left(\frac{1}{2} \right)^{4n} \right]}{1 - \frac{1}{2^n}}$

$= \frac{2^{4n} - 1}{2^{4n}\left(2^n - 1 \right)}$

38. ACD

Sol. $\frac{1}{x} + \frac{1}{y} = 1$

$\Rightarrow x + y = xy$

$\Rightarrow (x-1)(y-1) = 1$

$\Rightarrow x = 2, \quad y = 2$

39. BD

Sol. Let $t_r = \left(\frac{1}{r(r+1)} \right)^3 = \left(\frac{1}{r} - \frac{1}{r+1} \right)^3 = \frac{1}{r^3} - \frac{1}{(r+1)^3} - \frac{3}{r(r+1)} \left(\frac{1}{r(r+1)} \right)$

$= \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3\left(\frac{(r+1)-r}{r(r+1)} \right)^2, \quad t_r = \frac{1}{r^3} - \frac{1}{(r+1)^3} - 3\left(\frac{1}{r^2} + \frac{1}{(r+1)^2} \right) + 6\left(\frac{1}{r} - \frac{1}{r+1} \right)$

$\sum_{r=1}^{n} t_r = \left(1 - \frac{1}{(n+1)^3} \right) - 3\left(\frac{\pi^2}{6} + \frac{\pi^2}{6} - 1 \right) + 6\left(1 - \frac{1}{n+1} \right)$

$\lim_{n \to \infty} \sum t_r = 10 - \pi^2$

40. ABC

Sol. $R=\left(1,\frac{1+3\lambda}{1+\lambda}\right), \quad \therefore\left(\frac{1+3\lambda}{1+\lambda}\right)^2-4<0 \qquad \Rightarrow \lambda\in\left(-\frac{3}{5},1\right)$

41. ABC

Sol. Since $\sec^2\theta>1$

$\Rightarrow\left[(n+1)\sec^2\theta\right]>\left[n\sec^2\theta\right]$

Hence, f *and* g are both one-one

Let $k<n\sec^2\theta<k+1 and\ k<m\cos ec^2\theta<k+1$

$k\cos^2\theta<n<(k+1)\cos^2\theta\ and\ k\sin^2\theta<m<(k+1)\sin^2\theta$

Adding gives $k<\text{integer}<k+1$

$\Rightarrow A\cap B=\phi$

Suppose $k\notin A$

$\Rightarrow n\sec^2\theta<k\ and\ (k+1)<(n+1)\sec^2\theta$

$\Rightarrow n<k\cos^2\theta and\ n+1>(k+1)\cos^2\theta$

$\Rightarrow k<(k-n)\cos ec^2\theta, k+1>(k-n)\cos ec^2\theta$

$\Rightarrow\left[(k-n)\cos ec^2\theta\right]=k$

$\Rightarrow k\in B$

42. BCD

Sol. Orthocentre of ΔPQR lies on hyperbola

$\Rightarrow\lambda=3$

Incentre of ΔABC is orthocentre of ΔPQR

Equation of AB is $x+y-\frac{26}{5}=0$

43. AB

Sol. $(3\sec x-2)+3(2\sec x-1)^{\frac{1}{3}}(\sec x-1)^{\frac{1}{3}}=1$

$\Rightarrow(2\sec x-1)^{\frac{1}{3}}(\sec x-1)^{\frac{1}{3}}=(1-\sec x)$

$(2\sec x-1)^{\frac{1}{3}}=-(\sec x-1)^{\frac{2}{3}}\ or \sec x-1=0$

$\Rightarrow x=2n\pi, n\in 1$

44. ABCD

45. C

46. B

47. D

48. B

SECTION – C

49. 00024.00

Sol. $l = \ln \lim_{t\to 0} \dfrac{\int_0^t (1+2\sin 3x)^{4/x}\,dx}{t}$

$= \ln \lim_{t\to 0} (1+2\sin 3t)^{4/t}$ (Using L' Hospital's rule)

$= \ln e^{\lim_{t\to 0} \frac{4}{t}(2\sin 3t)}$

$= \lim_{t\to 0} \dfrac{2.3.4.\sin 3t}{3t} = 24$

50. 00002.00

51. 00000.00

Sol. $(x+1)^3 + (y+1)^3 = 16$

$y = -1 + \left(16 - (x+1)^3\right)^{1/3} = f(x)$

$x = -1 + \left(16 - (y+1)^3\right)^{1/3}$

$\therefore f^{-1}(x) = -1 + \left(16 - (x+1)^3\right)^{1/3} = g(x)$

$\Rightarrow f(x) = g(x)$

$f\left(-1+\sqrt[3]{15}\right) = g\left(-1+\sqrt[3]{15}\right) = 0$

$\left(f(x+g(x)).g(x+f(x))\right) = \left(f(x+f(x))\right)^2$

$\dfrac{d}{dx}\left(f(x+f(x))\right)^2 = 2f(x+f(x))$

$f'(x+f(x)).(x+f'(x))$

$\dfrac{d}{dx}\left(f(x+f(x))\right)^2 \; at \; x = -1+\sqrt[3]{15} = 0$

52. 00007.70

Sol. $f(x) = x - \sin x$

$f'(x) = 1 - \cos x$

$f'(x) = 0 \Rightarrow 1 - \cos x = 0$

$\Rightarrow \cos x = 1$

$x = 0, 2\pi$

$y = 0 \; and \; y = 2$ are two parallel tangents and distance between them $l_1 = 2\pi$

Also other two parallel tangents are parallel to the line $y = x$

$f'(x) = 1 \Rightarrow 1 - \cos x = 1 \quad \Rightarrow \cos x = 0 \quad \Rightarrow x = \dfrac{\pi}{2}, \dfrac{3\pi}{2}$

Tangent at $x=\frac{\pi}{2}$

$$y-\left(\frac{\pi}{2}-1\right)=1\left(x-\frac{\pi}{2}\right)$$

$$x-y=1$$

$$l_2=2\left|\frac{1}{\sqrt{2}}\right|=\sqrt{2}$$

$$\therefore l_1+l_2=2\pi+\sqrt{2}=7.70$$

53. 00002.14

Sol.
$$\sum_{n=0}^{\infty}\tan^{-1}\left(\frac{\tan^{-1}\left(\frac{(n+2)-(n+1)}{1+(n+1)(n+2)}\right)}{1+\tan^{-1}\left(\frac{1}{n+1}\right)\tan^{-1}\left(\frac{1}{n+2}\right)}\right)$$

$$=\sum_{n=0}^{\infty}\tan^{-1}\left(\frac{\tan^{-1}\left(\frac{1}{n+1}\right)-\tan^{-1}\left(\frac{1}{n+2}\right)}{1+\tan^{-1}\left(\frac{1}{n+1}\right).\tan^{-1}\left(\frac{1}{n+2}\right)}\right)$$

$$=\sum_{n=0}^{\infty}\left(\tan^{-1}\left(\frac{1}{n+1}\right)\right)-\tan^{-1}\left(\tan^{-1}\left(\frac{1}{n+2}\right)\right)$$

$$=\tan^{-1}\left(\tan^{-1}(1)\right)=\tan^{-1}\left(\frac{\pi}{4}\right)$$

$$\therefore p=\pi$$

$$\cos^{-1}\left(\cos(\pi-1)\right)=\pi-1=2.14$$

54. 00002.24

Sol. Since $z_1+z_2+z_3=1$

$$\Rightarrow \overline{z_1+z_2+z_3}=1$$

$$\Rightarrow \frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}=1 \qquad \Rightarrow z_2z_3+z_3z_1+z_1z_2=1$$

Now, consider a cubic equation having roots z_1, z_2, z_3

$$(z-z_1)(z-z_2)(z-z_3)=z^3-(z_1+z_2+z_3)z^2+(z_1z_2+z_2z_3+z_3z_1)z-z_1z_2z_3=0$$

$$\Rightarrow z^3-z^2+z-1=0$$

$$\Rightarrow z^2(z-1)+1(z-1)=0$$

$$\Rightarrow z=1, z=\pm i$$

$$\therefore \text{Roots } z_1=-i \;\; z_2=1 \;\; z_3=i$$

$$\left|z_1+z_2^2+z_3^3\right|=\sqrt{5}$$